# Literature Status: Weak-Star Derived-Set Questions from arXiv:0905.2531

agent_lane_14

2026-07-18

## Status

**Status: literature already answered.** The weak-star derived-set questions in Garcia–Kalenda–Maestre, *Envelopes of open sets and extending holomorphic functions on dual Banach spaces*, arXiv:0905.2531, Questions 6.3(a) and 6.5, are explicitly answered by M. I. Ostrovskii, *Weak$^*$ closures and derived sets in dual Banach spaces*, arXiv:1003.5176.

## Source Questions

On page 19, the source paper asks:

> Question 6.3(a). Let $X$ be a quasireflexive Banach space. Is $\overline{A}^{w^*} = A^{(1)}$ for each (absolutely) convex set $A \subset X^*$?
>
> Question 6.5. For which Banach spaces $X$ is there a linear subspace $A \subset X^*$ such that $A^{(1)}$ is a proper norm-dense subset of $X^*$? Is it true whenever $X$ is not quasireflexive?

We continue by questions on convex sets.

**Question 6.3.** *Let $X$ be a quasireflexive Banach space.*

(a) *Is $\overline{A}^{w^*} = A^{(1)}$ for each (absolutely) convex set $A \subset X^*$?*
(b) *Is each (absolutely) convex open subset of $X^*$ boundedly regular?*

Note that the question (a) has positive answer if $A$ is a linear subspace. Anyway the respective proof strongly uses linearity and it seems not to be clear how to adapt it to the (absolutely) convex case.

We also remark that both questions have positive answer if $X$ is reflexive. Indeed, in this case weak* topology on $X^*$ conincides with the weak one. Moreover, the weak closure of any convex set equals to its norm closure. So, $\tilde{U} = U$ for each open convex set $U \subset X^*$.

**Question 6.4.** *Let $X$ be a Banach space, $A \subset X^*$ an (absolutely) convex open set. Is it true that $A + rB_{X^*}$ is boundedly regular for each $r > 0$ if and only if $\overline{A}^{w^*} = A^{(1)}$?*

Proposition 4.5 shows that it is true if $A$ is a weak* dense linear subspace.

**Question 6.5.** *For which Banach spaces $X$ there is a linear subspace $A \subset X^*$ such that $A^{(1)}$ is a proper norm-dense subset of $X^*$? Is it true whenever $X$ is not quasireflexive?*

Note that we have just one example of such a subspace for $X = c_0$. It seems not to be clear how to adapt it even for $X = \ell_1$.


### Acknowledgement

We are grateful to Jiří Spurný for helpful discussions on the subject.


## Supporting Answer

Ostrovskii's introduction explicitly says that the paper answers these two questions from Garcia–Kalenda–Maestre. The stated main theorems give the answer:

- Theorem 1 answers Question 6.5 exactly: such a linear subspace exists if and only if $X$ is non-quasireflexive and contains an infinite-dimensional subspace with separable dual.
- Theorem 3 answers the absolutely convex part of Question 6.3(a) affirmatively: if $X$ is quasireflexive and $A \subset X^*$ is absolutely convex, then $A^{(1)} = \overline{A}^{w^*}$.
- Theorem 2 shows that the unrestricted convex version is false in every non-reflexive Banach space, hence in non-reflexive quasireflexive spaces.

Recently derived sets were used in the study of extension problems for holomorphic functions on dual Banach spaces [8].

The main purpose of the present paper is to answer the following two questions asked in [8]:

**1.** [8, Question 6.3(a)]. Let $X$ be a quasi-reflexive Banach space. Is $\overline{A}^{*} = A^{(1)}$ for each (absolutely) convex set $A \subset X^*$?

**2.** [8, Question 6.5]. For which Banach spaces $X$ there is a linear subspace $A \subset X^*$ such that $A^{(1)}$ is a proper norm-dense subset of $X^*$? Is it true whenever $X$ is not quasi-reflexive?

The main results of the paper:

**Theorem 1.** *The dual Banach space $X^*$ contains a linear subspace $A \subset X^*$ such that $A^{(1)}$ is a proper norm-dense subset of $X^*$ if and only if $X$ is a non-quasi-reflexive Banach space containing an infinite-dimensional subspace with separable dual.*

**Theorem 2.** *Let $X$ be a non-reflexive Banach space. Then there exists a convex subset $A \subset X^*$ such that $A^{(1)} \neq \overline{A}^{*}$.*

**Theorem 3.** *Let $X$ be a quasi-reflexive Banach space and $A \subset X^*$ be an absolutely convex subset. Then $A^{(1)} = \overline{A}^{*}$.*

Some parts of Theorems 1 and 2 are proved for separable spaces with basic sequences of special kinds first, and then are extended to the general case.

To describe the way in which results are extended from subspaces we need some more notation. Let $Z$ be a subspace in a Banach space $X$ and $E : Z \to X$ be the natural isometric embedding. Then $E^* : X^* \to Z^*$ is a quotient mapping which maps each

## Scope Limitations

This packet records only the literature answer to Questions 6.3(a) and 6.5. It does not resolve Question 6.2 about balanced sets, Question 6.3(b) about bounded regularity of open convex sets in quasireflexive duals, or Question 6.4 about the sets $A + rB_{X^*}$.

## References


- D. Garcia, O. F. K. Kalenda, and M. Maestre, *Envelopes of open sets and extending holomorphic functions on dual Banach spaces*, J. Math. Anal. Appl. 363 (2010), 663–678, arXiv:0905.2531.

- M. I. Ostrovskii, *Weak* closures and derived sets in dual Banach spaces*, Note Mat. 31 (2011), 129–138, arXiv:1003.5176.